

PDF-3: Executable Witnesses & Diagnostic Closure

Abstract

PDF-3 is a closure and registry artifact for diagnostic witness labels used across the frozen KEA artifact chain (PDF-0/1/2). It introduces no new semantics, predicates, metrics, thresholds, or execution logic. Its sole function is structural: (i) to anchor a minimal witness label set to inherited phase classifications, (ii) to provide explicit dependency closure to upstream immutable artifacts, and (iii) to state refutation conditions for witness registration under inherited semantics (including STOP as a terminal epistemic boundary).

1 Artifact Class

This document is classified as an *Executable Witness Registry and Diagnostic Closure Artifact*.

Artifact Nature. PDF-3 is a *registry-level closure artifact*. It introduces no new semantics, diagnostic logic, predicates, metrics, thresholds, evaluative procedures, or interpretations.

All diagnostic meaning, executability, and formal validity are *strictly inherited* from a frozen set of upstream artifacts (PDF-0, PDF-1, PDF-2). PDF-3 neither amends nor supplements those semantics.

Meaning of “Executable”. The qualifier *Executable* applies exclusively to diagnostic bindings, procedures, and evaluation mechanisms defined in upstream artifacts (in particular PDF-1).

PDF-3 itself contains no runnable components, algorithms, implementations, or execution logic, and does not claim executability as an isolated artifact.

Executability in PDF-3 is therefore *referential*: it denotes explicit attachment to inherited executable semantics, not intrinsic operational capability.

Interpretation Boundary (Executable). The term *Executable* has no descriptive content at the level of PDF-3 itself. Any interpretation of this artifact as participating in execution, validation, or evaluation—directly or indirectly—constitutes a category error relative to its declared artifact class.

Primary Function. The sole function of PDF-3 is to provide a formally closed registry that:

- fixes a canonical and finite set of diagnostic witness labels,
- anchors each witness label to an inherited diagnostic class or outcome,
- specifies explicit refutation conditions for witness registration,

- declares dependency closure over a frozen upstream artifact set.

PDF-3 performs no evaluation and authorizes no execution. It records diagnostic closure only.

Diagnostic Closure Invariant. Upon publication of PDF-3, the following invariant holds:

All admissible diagnostic outcomes under the inherited semantics are exhaustively labeled, mutually exclusive, and refutable. No additional witness category can be introduced without modifying upstream semantics.

This invariant constitutes the diagnostic closure provided by PDF-3.

Non-Semantic Scope. PDF-3 operates purely at the structural and bookkeeping level. It fixes classification boundaries, label admissibility, and dependency constraints, but performs no inference, computation, execution, or validation.

Any future executable realization, empirical instantiation, or formal proof must reference the witness labels defined here without altering their inherited meaning.

1.1 Structurally Fixed Elements (Non-Semantic)

Relative to the inherited semantics, PDF-3 irreversibly fixes:

- a minimal and canonical witness label set for diagnostic closure,
- a total mapping between witness labels and inherited diagnostic classes,
- an explicit dependency-closure declaration over frozen upstream artifacts,
- explicit refutation conditions specifying when a witness label is unjustified.

2 Scope and Positioning

Scope Definition. PDF-3 operates exclusively as a *diagnostic closure and registry layer* within a fixed artifact chain. Its scope is strictly limited to referencing, labeling, and closing diagnostic outcomes defined elsewhere.

PDF-3 depends *only and entirely* on a frozen set of upstream artifacts. It introduces no independent semantics, diagnostics, execution logic, interpretive authority, or evaluative criteria of its own.

Complete Upstream Dependency Set. All content in PDF-3 is defined relative to the following upstream artifacts:

- PDF-0: KEA Whitepaper (foundational formal specification),
- PDF-1: Executable Demonstration Artifact (executable bindings and evaluation procedures),
- PDF-2: Structural Diagnosability & Phase Logic (diagnostic phase semantics).

No other documents, implementations, datasets, or interpretations are within scope for this artifact.

Non-Extension Guarantee. PDF-3 does not modify, reinterpret, extend, refine, operationalize, or contextualize any content originating in the upstream artifacts.

All semantics, classifications, predicates, thresholds, phase definitions, and diagnostic meanings are inherited *verbatim and without transformation*. Any semantic effect observable in PDF-3 is reducible to upstream definitions.

Position Within the Artifact Chain. PDF-3 occupies a strictly terminal position in the artifact sequence. Its role is limited to:

- anchoring witness labels to inherited diagnostic classifications,
- declaring explicit refutation conditions for those labels,
- formally closing the diagnostic classification space defined upstream.

PDF-3 performs no inference, evaluation, execution, or validation. It records diagnostic closure only.

Non-Autonomy Constraint. PDF-3 is *intentionally non-autonomous*. Its validity, meaning, and interpretability are entirely derivative:

In the absence of the frozen upstream artifacts, PDF-3 has no semantic content, no diagnostic meaning, and no executable interpretation.

This non-autonomy is not a limitation but a defining structural property of the artifact class.

2.1 Dependency Closure (Frozen Set)

Frozen Dependency Set. PDF-3 is defined only relative to an immutable dependency set. The following upstream artifacts are treated as frozen inputs for all inherited semantics referenced in this document:

Dependency	Frozen Reference (Informational Anchor)
PDF-0	KEA Whitepaper (foundational formal specification)
PDF-1	Executable Demonstration Artifact (execution bindings and evaluation procedures)
PDF-2	Structural Diagnosability & Phase Logic (diagnostic phase semantics)
SHA-256 anchor	c2385be94c79f52c9cbe6ee4437994c8833224cdce50ed8665241c0fdb7ecce6

Closure Validity Condition. The dependency set above constitutes a *validity boundary* for PDF-3. Any modification, replacement, extension, or reinterpretation of the upstream artifacts invalidates this document as a diagnostic closure artifact.

Integrity Handle (Recorded, Not Verified). The SHA-256 anchor is recorded solely as an external integrity handle identifying the frozen chain state used in the current run. PDF-3 performs no verification and asserts no trust mechanism; it records the dependency closure reference only.

3 Non-Claims

Explicit Non-Claims. PDF-3 explicitly makes no claims of any kind regarding:

- correctness, completeness, sufficiency, robustness, or optimality,
- applicability to any specific organization, system, domain, or context,
- predictive, prescriptive, normative, advisory, or optimization outcomes,
- empirical validation, performance characteristics, or effectiveness,
- success, failure, or quality of any execution.

Structural Nature of Non-Claims. The absence of such claims is not incidental or defensive. It is a *structural property* of the artifact class instantiated by PDF-3.

PDF-3 does not evaluate systems, executions, models, organizations, or decisions. It neither asserts nor implies desirable outcomes, recommended actions, or performance expectations. Its function is strictly limited to the formal closure and registration of diagnostic classifications inherited from upstream artifacts.

Interpretation Boundary. Any interpretation of this document as:

- an evaluation of a system,
- an assessment of organizational capability,
- a recommendation, guideline, or method,
- or an operational or managerial instrument

constitutes a *category error* relative to the declared artifact class.

Such interpretations are invalid independently of whether they appear plausible, useful, or aligned with external goals.

Separation of Concerns. All claims concerning empirical behavior, operational effectiveness, organizational outcomes, or decision quality—if made at all—belong exclusively to contexts external to PDF-3. They are neither supported, implied, nor contradicted by this document.

PDF-3 terminates diagnostic meaning; it does not extend it.

4 High-Level Structure

Structural Role. PDF-3 is not organized as a narrative, method description, or executable module. Its structure follows directly from its declared artifact class: a diagnostic closure and registry layer positioned above executable and semantic artifacts.

As such, the structure of PDF-3 is intentionally minimal and non-creative. It exists to make explicit what is already fixed elsewhere.

Constituent Structural Elements. The artifact consists exclusively of the following structural components:

- a canonical registry of diagnostic witness labels used for closure bookkeeping,
- explicit references from each witness label to its originating diagnostic predicates and phase classifications defined in upstream artifacts,
- closure markers that bind each witness label to a frozen dependency set (PDF-0, PDF-1, PDF-2).

No additional structural elements are permitted without violating the declared scope and role of the artifact.

Absence of Execution by Construction. No witnesses are implemented, instantiated, or executed within PDF-3. This absence is not a limitation or omission. It is a direct consequence of the artifact’s position in the document chain.

All executable bindings, operational procedures, and evaluation logic reside exclusively in upstream artifacts (in particular PDF-1). Embedding or duplicating execution logic here would constitute a structural error, resulting in semantic ambiguity and invalidating dependency closure.

Closure Invariant. PDF-3 satisfies the following invariant:

Every diagnostic label appearing in this artifact is fully determined by, and referentially reducible to, frozen upstream semantics.

The artifact introduces no new axes of variation, interpretation, or execution. Its structure is exhausted by registry, reference, and closure.

5 Diagnostic Witnesses (Non-Executable)

Role of Witnesses in PDF-3. In PDF-3, a *diagnostic witness* is neither an execution trace, nor an algorithm, proof, or empirical observation. It is a *structural closure token* that certifies the outcome of an evaluation performed entirely in upstream artifacts.

Witnesses exist to make diagnostic outcomes:

- referentially explicit,
- unambiguous across documents,
- stable under frozen dependencies.

They provide a formally fixed interface between executable diagnostics and downstream interpretation, without reintroducing execution or semantics.

Non-Executability by Construction. All witnesses defined in this section are intentionally non-executable. This is not an omission but a structural necessity imposed by the artifact hierarchy.

Evaluation, execution, and classification occur exclusively in upstream artifacts. PDF-3 records only the diagnostic closure of those evaluations.

Any executable content at this level would duplicate or override inherited semantics, violating dependency closure and invalidating the artifact class.

Hard Semantic Constraint. All semantics used to justify witness registration are inherited *verbatim and without modification* from the frozen upstream artifacts: PDF-0 (KEA Whitepaper), PDF-1 (Executable Demonstration Artifact), and PDF-2 (Structural Diagnosability & Phase Logic).

PDF-3 introduces no predicates, thresholds, metrics, evaluative procedures, or formal conditions of its own.

Epistemic Discipline. All witnesses are strictly:

- diagnostic-only,
- non-optimizing,
- non-prescriptive.

No witness implies control, remediation, intervention, optimization, or normative judgment beyond inherited diagnostic classification.

5.1 Canonical Witness Set (Closure Completeness)

PDF-3 fixes a minimal, finite, and sufficient canonical set of diagnostic witness labels.

Witness	Target Class	Structural Diagnostic Meaning
W_S	SUCCESS	Inherited evaluation yields SUCCESS under preserved diagnosability.
W_I	INERTIA	Inherited evaluation yields INERTIA: bounded, persistent diagnosability without structural progression.
W_C	COLLAPSE	Inherited evaluation yields COLLAPSE: irreversible loss of structural viability.
W_{STOP}	STOP	Inherited diagnosability conditions fail; evaluation terminates without classification.

Structural Scope of Completeness. The terms *minimal*, *finite*, and *sufficient* refer exclusively to structural closure relative to the frozen upstream semantics. They do not assert a theorem, proof, optimality, or completeness property independent of those semantics.

Closure Completeness Statement. This witness set is *exhaustive* with respect to inherited semantics:

For every admissible evaluation outcome under the frozen upstream semantics,
there exists a witness label in this set that is registrable in PDF-3.

No additional witness category can be introduced without modifying upstream diagnostic logic.

5.2 Witness Registration Principle

A witness label W_X is registrable in PDF-3 if and only if:

An evaluation performed under frozen upstream semantics (PDF-1 and PDF-2) yields the diagnostic class X , and no violation of diagnosability invalidates that evaluation.

PDF-3 neither defines nor executes this evaluation. It records only the resulting closure label.

5.3 Witness W_S : SUCCESS

Diagnostic role. Registers that a referenced configuration is classified as SUCCESS under inherited semantics.

Conceptual input (non-executable). A phase-space configuration admissible under the frozen constraints of PDF-1, evaluated using the phase logic defined in PDF-2.

Closure condition. The inherited classifier yields SUCCESS and diagnosability is preserved.

Refutation (inherited). W_S is invalid if:

- the inherited classifier does not yield SUCCESS, or
- diagnosability is lost and STOP is triggered.

5.4 Witness W_I : INERTIA

Diagnostic role. Registers that a referenced configuration is classified as INERTIA under inherited semantics.

Conceptual input (non-executable). A phase-space configuration remaining admissible under frozen constraints while yielding the INERTIA classification.

Closure condition. The inherited classifier yields INERTIA without loss of diagnosability.

Refutation (inherited). W_I is invalid if:

- the classifier yields SUCCESS, COLLAPSE, or STOP, or
- admissibility conditions are violated.

5.5 Witness W_C : COLLAPSE

Diagnostic role. Registers irreversible loss of structural viability classified as COLLAPSE under inherited semantics.

Conceptual input (non-executable). A configuration that, when evaluated under frozen upstream semantics, yields COLLAPSE.

Closure condition. Immediate classification as COLLAPSE.

Refutation (inherited). W_C is invalid if the inherited classifier does not yield COLLAPSE.

5.6 Witness W_{STOP} : STOP

Diagnostic role. Registers terminal epistemic non-diagnosability.

Conceptual input (non-executable). A configuration for which inherited diagnosability conditions are not satisfied.

Closure condition. Immediate termination of evaluation via STOP.

STOP Axiom (Terminality). STOP is a *terminal epistemic boundary condition*. It is not a failure, error, defect, or system state. No diagnostic, evaluative, inferential, or prescriptive operation is permitted beyond STOP.

Non-Failure Interpretation. STOP is not a failure mode, error condition, defect, or system state. It denotes a terminal epistemic boundary beyond which no diagnostic, evaluative, inferential, or prescriptive interpretation is defined or permitted.

Refutation (inherited). W_{STOP} is invalid if diagnosability remains satisfied under frozen upstream semantics.

5.7 Non-Prescription Invariant

Witness registration in PDF-3 has no operational consequences. It neither implies nor authorizes action, intervention, optimization, or decision-making.

The role of PDF-3 is exhausted by diagnostic closure.

6 Closure Note

This artifact establishes and fixes the artifact class, scope boundaries, and diagnostic closure role of PDF-3 within the frozen document chain.

No additional semantics, executables, evidence, or formal content are introduced in this run.

The function of PDF-3 is exhausted by the explicit registration and closure of inherited diagnostic classifications and their refutation conditions under inherited semantics.

External Review Notes / Reviewer Disposition

PDF-3 has been subjected to external neural review for the purpose of identifying misinterpretations, category errors, and boundary violations relative to its declared artifact class.

External reviewers raised requests including executable instantiation, worked examples, formal proofs, bibliographic positioning, and verification of dependency hashes. These requests are not deficiencies of the artifact. They constitute category errors relative to the declared role of PDF-3 as a non-autonomous diagnostic closure and registry layer.

PDF-3 is intentionally non-executable and non-semantic. All semantics, executability, validation, and formal meaning are inherited exclusively from a frozen set of upstream artifacts. No such requests are satisfied in this document by design.